

XTemp-NLP: Planting Knowledge Conflicts into LLM. Eliciting and Analysing Approaches for Targeted Cases

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Abstract

Hallucinations and the inconsistency of Large Language Models (LLMs) prevent the automation of language- and knowledge-intensive tasks in domains that require the utmost precision, such as medicine. One of the main reasons being intra-memory knowledge conflicts that models acquire during pre-training. We aim to explore their emergence in the medical domain. To this end, our approach injects perturbed data by synthesizing conflicting data into training. Evaluation on a downstream task shows model develop conflicting representations with successive epochs. We find that introducing conflicts negatively affects language understanding and results in slightly lower performance on the QA task. We make our implementation fully available on GitHub¹.

1 Introduction

Large Language Models (LLMs) are currently used in a variety of NLP and reasoning tasks, from summarisation to fact-checking, and replacing search engines. In their performance, they rely on factual knowledge, memorised during pre-training and stored as parametric knowledge in the model (Yu et al., 2024). However, due to the enormous scale of pre-training data, this knowledge is prone to contain misinformation, biases, and outdated facts, not seldom causing the model to produce factually incorrect outputs (Pan et al., 2023).

While humans are generally able to identify such conflicting information, *consolidate their memory*, and assign judgment labels to statements, models lack this mechanism and struggle with continuous learning, failing to take different views into account.

LLMs showing remarkable capabilities in healthcare, with prominent applications such as clinical

text comprehension and reasoning, and even showcasing physician-level performances (Singhal et al., 2023a), their successful clinical integration still hinges on rigorously understanding their limitations. One of which is the ability to adapt to constantly evolving knowledge. We aim to contribute to this under-studied field through approaches targeting eliciting (1) *how models store conflicts* (during pre-training) and (2) *how conflicts manifest* (in a downstream task).

This project focuses on knowledge conflicts in LLMs. Precisely, our approach aims to study intra-memory conflicts in a controlled manner to analyse how they emerge during training. We hypothesise that models pre-trained on *conflicting* data, i.e. deliberately adjusted contexts that inherently conflict with original versions, will show consistent case evidence on the downstream task.

For our experiments, we deploy a novel sparse-attention transformer architecture, BigBird, designed to scale efficiently to long sequences using sparse attention, while retaining the expressiveness of full-attention mechanisms. Other more specialised objectives have proposed, like SpanBERT (Joshi et al., 2020) or Recurrent Span Selection (RSP) (Lee et al., 2017), which follow different masking schemes (i.e. not only token-level). Finally and most importantly, BigBird is a general-domain model that was not trained on medical data, making it tabula rasa to the data we train it on.

Our methodology includes carefully curating *conflict-perturbed* data alongside original data for model pre-training, and a controlled evaluation setup for targeted analysis of *conflicts* on a downstream task.

Our main steps, having obtained the cleaned version of the MedMCQA dataset, are:

1. Generating conflicting explanations;
2. Generating multiple question paraphrases;
3. Training lightweight BigBird in two settings;

¹<https://github.com/bohdana-ivakhnenko/knowledge-conflicts>

conflict-free and with planted conflict;

We contribute to the MedMCQA dataset by augmenting it with plausible, modified contexts and multiple paraphrased questions.

2 Related Work

The phenomenon of LLMs over-relying on their parametric knowledge instead of adhering to facts not contained in the input, (Longpre et al., 2021) first term as "hallucinations".

They originate from knowledge conflicts, of which there are three main types depending on the source (Huang et al., 2023):

1. *Inter-context conflicts* originate from the model input during inference, especially from the context provided alongside the prompt. This mostly happens in retrieval-augmented generation (RAG).
2. *Memory-context conflicts* are easily described in temporal view. Once the model is trained, its weights cannot be updated further, so new information can only be channelled through prompt context. Conflict occurs when the model's knowledge becomes outdated.
3. *Intra-memory conflicts* originate from the training data. During pre-training, the model learns language probabilities while also learning to give conflicting predictions under superfluous differences in prior.

The pre-training data is recognised as the primary cause of *intra-memory conflicts* (Xu et al., 2024). Multiple syntactic representations of semantically identical statements arise from different sources, contributing to LLMs' uncertainty and instability in factual recall. As a result, LLMs answer differently when given the same prompt in different wording, such as "Who is the president of France?" and "The president of France is...", as they lead to different probability distributions over the final word.

At the moment, intra-memory conflicts are the least studied of the varieties. Existing papers focus on discovering (Xu et al., 2022), (Zhao et al., 2024), (Zhang et al., 2023), (Hase et al., 2023), but mostly on mitigating them by reducing bias in the training data (Wang et al., 2023), employing non-standard decoding strategies (Chuang et al., 2023), (Lee et al., 2022), and technical solutions based on shifting model activations (Li et al., 2023).

Besides models storing incorrect facts with a single valid representation, models may also need to

confine to multiple different versions of facts, often stemming from temporality, e.g. asking about the presidency from different timeframes for the above example. As such, (Marjanovic et al., 2024) distinguish between static vs dynamic facts, with *dynamic* referring to conflicting viewpoints and temporally sensitive data. Dynamic facts are especially typical to the medical field, due to the rapid expansion of biomedical knowledge driven by swift research and advancements (Densen, 2011). Termed medical *concept drift* (Wu et al., 2025a), clinical guidelines, the formal standards of medical knowledge constantly evolve, causing standard practices to quickly become obsolete (Chopra et al., 2023). In current efforts to develop machine-learning-based solutions in the medical domain, the ability of models to adapt becomes critical.

When training the model on scarce medical data, outdated guidelines inevitably create contradictions (Wu et al., 2025a). The NICE-SUGAR study ((NICE-SUGAR Study Investigators et al., 2009); (Cagnacci and Venier, 2019)) exemplifies this with treatment recommendations that were later found to increase mortality. As such, LLMs' ability to recognise and discern between apparent, sometimes subtle conflicts is critical for enabling trustworthiness and effectiveness of their clinical deployment and implementation.

Evaluation methods for assessing and mitigating intra-memory conflicts have been recently proposed. (Marjanovic et al., 2024) introduce DYNAMICQA, a benchmark containing retrieved realistic contexts as real-world knowledge. Furthermore, they manipulate these contexts by augmenting similar, competing objects and evaluating with two conflict measures, one of which is semantic entropy. For assessing how LLMs manage internal conflicts, (Wu et al., 2025a) propose ConflictMedQA, a benchmark that pairs each clinical current recommendation with a mutually exclusive, pseudo-outdated version. Through this, they demonstrate that models struggle to reject incorrect recommendations and explore retrieval-augmented generation and preference-finetuning as two mitigation strategies.

Transfer learning is an established method of improving the performance of target learners in target domains by transferring knowledge from related source domains of current machine learning approaches (Zhuang et al., 2020). Domain adaptation (Zhuang et al., 2020), as a distinct step in transfer

learning, describes reducing differences between domains and, as such, adapting source domains to transfer knowledge.

3 MedMCQA Dataset

We use the MedMCQA dataset as a basis for our synthetic dataset (Pal et al., 2022).

This is a human-curated, high-quality dataset of 194 thousand multiple-choice questions (MCQs) collected from the entrance exam to All India Institute of Medical Sciences (AIIMS) colleges and the National Eligibility cum Entrance Test for Postgraduates (NEET PG). The questions cover 20 medical subjects.

Each entry consists of a unique `id`, a question, answer options (and the index of the correct one), an explanation that justifies the correct answer, and the subject to which the question belongs. See an example in 1.

Even though the dataset is generated and reviewed by humans, we observe issues hindering its automatic processing in tasks requiring high precision:

1. No unified question-formulation style; i.e., many lack question words and question marks.
2. Not all entries contain explanations. Approximately half of `dev`, 12% of `train`, and all `test` entries lack them.
3. Existing explanations often reveal the correct answer option.
4. Some existing explanations are very short and do not provide enough information to arrive at the correct option.
5. Formatting issues with spacing and PDF OCR artefacts leading to spelling mistakes.
6. Duplicate entries and duplicate answer options.

We had a chance to use a cleaned version of the dataset (questions, explanations, and answer options) that addressed formatting issues. This version was produced automatically with a `meta-llama/Meta-Llama-3-8B-Instruct` model provided by Meta.

4 Methodology

This section presents and explains the design of our approach, its benefits, how we augmented the MedMCQA dataset, discusses the training approach, and concludes with the evaluation strategy employed.

Question: Which of the following is not true about glomerular capillaries?

Cleaned Question: Which of the following statements about glomerular capillaries is ****not**** true?

Options:

- A. The oncotic pressure of the fluid leaving the capillaries is less than that of fluid entering it
- B. Glucose concentration in the capillaries is the same as that in glomerular filtrate
- C. Constriction of afferent **aeiriole** decreases the blood flow to the **glomerulas**
- D. Hematocrit of the fluid leaving the capillaries is less than that of the fluid entering it

Cleaned Options:

- A. The oncotic pressure of the fluid leaving the capillaries is less than that of fluid entering it
 - B. Glucose concentration in the capillaries is the same as that in glomerular filtrate
 - C. Constriction of afferent **aeiriole** decreases the blood flow to the glomerulus
 - D. Hematocrit of the fluid leaving the capillaries is less than that of the fluid entering it
-

Explanation: **Ans-a.** The oncotic pressure of the fluid leaving the capillaries is less than that of fluid entering it **Guyton I LpJ1 4-;anong 23/e p653-6_)Glomerular oncotic pressure (due to plasma protein content) is higher than that of filtrate oncotic pressure in Bowman’s capsule\)**“Since glucose is freely filtered and the fluid in the Bowman’s capsule is isotonic with plasma, the concentration of glucose in the filtrate is the same as in the capillaries

Cleaned Explanation: The oncotic pressure of the fluid leaving the capillaries is less than that of fluid entering it. This is because glomerular oncotic pressure, which is influenced by the plasma protein content, is higher than the oncotic pressure of the filtrate in Bowman’s capsule. Since glucose is freely filtered, the concentration of glucose in the filtrate is the same as in the capillaries, and the fluid in the Bowman’s capsule is isotonic with plasma.

Correct Option: A

Subject: Physiology

ID: b944ada9-d776-4c2a-9180-3ae5f393f72d

Table 1: MedMCQA single-choice question example. The issues are coloured in orange.

4.1 Conflict Planting Approach

To investigate how conflicts arise in a model’s parametric memory, we must precisely control its pre-training data. Using a domain-specific model, such as one trained on medical data, would be unsuitable: such models already encode numerous facts and inconsistencies, making it difficult to attribute conflicts to controlled sources. Training a model from scratch was also impractical—both computationally prohibitive and infeasible given the labour required to assemble a sufficiently large domain corpus, especially in the medical domain.

Instead, we adopt a lightweight but effective approach: starting from a general-domain model with no prior medical knowledge. Such a model already

possesses linguistic competence while treating any medical input as new information, allowing us to deliberately introduce controlled knowledge conflicts. Since language models only encode what is present in their training corpus, any curated text collection can serve as a stand-in for “reality.”

To operationalise this setup, we require (1) a corpus suitable for pretraining and manipulation, and (2) a complementary QA dataset that exposes the induced conflicts. The two datasets must be topically aligned to enable careful tracing between the knowledge source and its behavioural manifestation. Most existing medical QA datasets targeting knowledge conflicts are inadequate for this purpose: they are typically small, lack raw textual contexts, or focus on open-ended, downstream evaluation. For example, ConflictMedQA (Wu et al., 2025b) lacks supporting texts; MedAlpaca (Han et al., 2023) and MedQuAD (Ben Abacha and Demner-Fushman, 2019) contain short, open-ended questions; and PubMedQA (Jin et al., 2019) is heavily imbalanced (“a trivial majority baseline achieves 92.76% accuracy”).

The MedMCQA dataset, by contrast, satisfies our requirements. It is large, human-authored, and links factual questions to detailed explanations. Moreover, the explanations provided in its cleaned extension can serve as stand-alone pretraining contexts, which we systematically perturb to construct conflicting evidence.

We generate original and misleading contexts for the same questions, combining them to create controlled contradictions. This synthetic corpus offers fine-grained control: subsets can be filtered by topic to evaluate models trained with or without misleading contexts. Finally, we paraphrase the questions to introduce syntactic variation, enabling a more robust examination of conflict retrieval in the QA setting.

The resulting setup establishes a fully controllable environment for conflict induction and analysis — forming the foundation for our Dataset Construction process described next.

4.2 Dataset Construction

4.2.1 Perturbed Explanations

We produce conflict-perturbed data to address the model’s handling of intra-memory knowledge conflicts. For this purpose, we use MedMCQA data and generate a plausible explanation justifying the incorrect answer for each sample. A sample d is

given as a tuple $d = (q, e, O, cop)$, where q denoting the question, explanation, the set of answer options $O = \{o_1, o_2, o_3, o_4\}$, along with the correct answer option $cop \in \{1, 2, 3, 4\}$, such that the correct answer is o_{cop} . Given a sample $d = (q, e, O, cop)$, we prompt the model to generate a synthesized pair $d_{mod} = (e_{mod}, cop_{mod})$, where e_{mod} is a modified explanation and $cop_{mod} \in \{1, 2, 3, 4\}$, $cop_{mod} \neq cop$. Thus, turning one of the remaining answer options correct for this instance. To obtain cop_{mod} , we parse the model answer to extract its corresponding index position in the given answer options. If the output cannot be matched to any option, we label the sample as *invalid* and record approximately 30% of such cases. We obtain a total of 9,881 valid modified explanations, approximately 1/3 of the original training data size (80k).

4.2.2 Question Augmentation

In order to test and surface intra-memory conflicts elicited through syntactic variance of the input prompt, we paraphrase the provided questions from the dataset’s split without providing additional context. We also decide to forgo the option of supplying the questions as part of the paraphrasing input, since it doubled the time per prompt without resulting in more faithful generations. We used Ollama’s² Llama3.2 3B model. Llama’s 3B checkpoint allowed for paraphrasing a question in 1.5 seconds while still yielding topically and grammatically correct results 92.35% of the time. From these 7.65% of suboptimal paraphrases, 2.3% were a result of badly formatted original questions, which then propagated through the model into the paraphrases. Thus, all of the original, but malformed questions, were removed to facilitate a higher-quality curated dataset. For examples on how these inputs look, please refer to Table ?? At the end, we obtain a 3.5 – 4 times increase in training data, reported in Table 2. The exact quality assurance undertaken for the paraphrasing can be found in the appendix section C. Overall, a mix of manual inspection and automated metrics was used to ensure that the paraphrases remained faithful and of high quality.

4.3 Training

Our method explores how a Transformer-based model can adapt to the target domain under

²<https://ollama.com/>

Split	Orig.	Aug.	Raw Inc.	Inc. %	Avg Gain/Entry
Train	182,822	812,550	629,728	344.5%	3.45
Valid	4,183	19,379	15,196	363.3%	3.63
Test	6,150	30,785	24,635	400.6%	4.01

Table 2: Comparison of original and augmented dataset sizes.

conflicting conditions. We use a lightweight `google/bigbird-roberta-base`³ model. *BigBird* is an English general-domain model, pre-trained on Books, CC-News (Wenzek et al., 2019), Stories, and Wikipedia, without any medical-domain data. It also performed *state-of-the-art* for tasks like long document summarization (Koh et al., 2022) and QA over long contexts (Tay et al., 2020).

Traditional BERT-based models (Devlin et al., 2019) use the standard attention mechanism introduced with the Transformer architecture (Vaswani et al., 2017). However, a well-known bottleneck of this method is the quadratic time and memory complexity with increasing sequence length, limiting the application due to the infeasible computational cost. (Zaheer et al., 2020) propose one of many effective solutions to the problem. They manage to extend the Transformer architecture to handle substantially longer sequences, up to 4096 tokens. To achieve this, they employ linear blocks of sparse attention, along with a predefined small number of global tokens attending to all input tokens. The key difference is that the number of global tokens grows only at a constant rate proportional to the input sequence. In other words, efficiency for long sequences is retained but at much lower computational cost.

Hence, we choose BigBird as our model for its general domain, fast convergence on specific domain terminology and processing of long chunks, followed by QA testing to investigate the effects of adaptation on (1) high-quality, manually curated and cleaned data, and (2) conflicting data.

4.3.1 Domain Adaptation

Domain adaptation (DA) (Singhal et al., 2023b) is a special form of continuous learning within transfer learning (Zhuang et al., 2020) that serves our purpose by allowing for data distribution shifts between source and target domains while the task remains the same. We first expose the model to a source distribution and subsequently adapt it to a

deliberately corrupted target distribution, thereby allowing us to systematically probe its internal representations under controlled perturbations. Initialising from a pre-trained model such as BigBird allows us to inherit general linguistic and structural representations, providing a strong inductive bias and significantly accelerating convergence. Our specific BigBird is a faster extension of RoBERTa’s checkpoints, thereby guaranteeing the same sentencepiece vocabulary of 50358 tokens. This is important, since we use it to estimate corpus size and customise our QA evaluation, as described in section 4.3.3.

During domain adaptation, we update all **128,111,286 M** parameters and refrain from using techniques such as layer-freezing (Yang et al., 2025). This enables propagating gradients through the whole network. Furthermore, the small size and lightweight architecture of the model give us full control rather than relying on widespread techniques like low-rank adapters (Hu et al., 2021), but still provide limited control. Although updating all parameters is generally useful for supervised tasks with labelled examples, it nevertheless poses the risk of catastrophic forgetting (Kirkpatrick et al., 2017), leading to the loss of learned general knowledge. In an additional study, we demonstrate that our model does not experience catastrophic forgetting in Section 5.

4.3.2 Training Strategy

We choose the standard Masked Language Modelling (MLM) (Devlin et al., 2019) pretraining objective for randomly masking a set percentage of tokens (usually 15%) of the input sequence, which the model then predicts.

Data scarcity poses a significant and long-standing challenge for deep learning across many domains, of which the medical field is not exempt. Using the original (cleaned) MedMCQA data and augmenting it with conflict-perturbed cases, as previously described, we report the following statistics. Table 3 gives an estimate of the tokens available for DA with all data collections. The corpora are restricted by the size of the conflicting contexts, as their number remains limited for this work. However, in general, corrupted contexts tend to be longer, which explains the 1.2M token difference between the baseline training dataset (no sampling) and the conflicting corpus dataset. That is why we decided to upsample the human-annotated data to ensure comparable sizes. The sampled entries are

³<https://huggingface.co/google/bigbird-roberta-base>

Dataset	Split	Size
Baseline Corpus	Train (w/o sampling)	2.147M
	Train (w/ sampling)	3.346M
Conflicting Corpus	Train	3.432M
Dev		0.2603M

Table 3: Corpora sizes used for baseline and conflicts experiments

chosen randomly, provided they do not overlap.

4.3.3 Evaluation Strategy

As it is rather irrelevant and out of scope to further fine-tune models for the downstream multiple-choice classification task, a more fine-grained evaluation method that shows the models’ decision-making process is better suited. Exactly for this reason evaluating our adapted models poses the biggest technical challenge. This is due to (1) we need to understand what the outcomes are and (2), we need to implement the evaluation loop that fits our needs. To address these concerns, we proceed by reformatting the QA task as a language modeling task. The full proposed pipeline is visualised in Figure 3. The following summarises our concrete steps:

- Each data sample is transformed as a product of possible combinations of the provided question and the answer options, yielding 4 question-answer paired tuples.
- Further, question and answer are separated by the [SEP] token to mark their boundaries (as a signal to the model).
- As such, we preserve the original MLM training objective of masking 15% of the tokens to be predicted.
- Each masked sequence is given to our adapted model, which returns a tensor with logprobs over the full vocabulary. Summing over the log probabilities of all tokens, we estimate the average likelihood of each question-answer paired sequence.
- Consequently, one out of four sequences that is assigned the highest probability, thus favoured by the model, we regard the model as judging it correct.
- Finally, we do the same for the best paraphrase of the question. More on paraphrasing in Section 4.2.2.

An answer is considered chosen by the model, iff the original (cleaned and updated) question and the

paraphrased version yield matching model answers. In cases of missing paraphrases, we simply repeat the original question twice, check for consistency, and note that this accounts for less than one-third of all instances. We always select a single paraphrase from the pool of paraphrased questions for each data entry, based on the highest BERTScore (Zhang et al., 2020), since data entries differ in the number of paraphrased questions.

Batching masked sequences allows us to quantify the effect of DA on the QA task directly, without further fine-tuning. We measure performance across several dimensions, such as training loss (and perplexity), evaluation loss, and masked token accuracy. Additionally, we evaluate on the original QA dataset and track accuracy per topic (the medical topics in the MedMCQA dataset), as well as macro precision, recall, and F1-score.

5 Experiments & Results

This section presents the different experimental designs and their results. With access to model weights from Hugging Face (HF)⁴, we develop all of our methods using PyTorch⁵ and the Transformers library.

There are three distinct experimental setups to better assess the impact of conflicting information while accounting for the models’ learning trajectories during training. The experiments are as follows:

1. **Baseline** - the baseline experiment uses all clean contexts from the human-curated train dataset, for which planted conflicts are also available. Since the baseline dataset is conflict-free, it could be half as large, so we avoid this by upsampling it with additional cases from the corpus described in section 4.3.2 to ensure comparability with other experiments.
2. **Conflicting** - in the conflicting experiment, we train on all corrupted contexts and their clean counterparts from the original train dataset. All concatenated into a single data collection.
3. **Off-the-shelf** - the off-the-shelf experiment is only for inference time, because we directly take the weights from the originally pretrained model. The purpose of this is to detect possible catastrophic forgetting, discussed in Sec-

⁴<https://huggingface.co/>

⁵<https://pytorch.org/>

tion 4.3.1.

We initiate all model variants with BigBird’s pretrained weights. The models are trained with identical hyperparameter settings to ensure fairness. The exact configurations are provided in the Appendix B.1. We try to adapt our models more subtly and use an initial learning rate of $3e-5$, linearly scheduling with warmup. The device training batch size is set to 4 and the gradient accumulation steps to 2, so the effective batch size is 8, with a final eval batch size of 4. We choose to leverage BigBird’s long input sequence length and concatenate all input explanations across batches of 2048 tokens. All models run for 8 full iterations on their respective train datasets. We do not use any early stopping callbacks or plateaus. All processes run on a single A100 GPU with 80 GBs of RAM. We adopt the standard practice of dynamically masking 15% of the input tokens per batch.

During training, we use the same dev set for all models, which consist of previously unseen text from gold-standard contexts. On each epoch, we additionally run the evaluation from Section 4.3.3 on our QA dataset. The QA dataset is formed by taking the questions from our conflicting dataset (which also appear in the baseline) and, for each question, taking the best paraphrase. The off-the-shelf experiment is only evaluated on this QA task.

First, we inspect how the **baseline** experiment compares to the **conflicting**. The results are visible in Table 4 and in Table 5. The final training loss for the baseline experiment is 3.8061, while it is 4.1005 for the conflicting experiment. This cross-entropy loss can be conveniently turned to perplexity by taking the exponential. One can think of perplexity as the number of other plausible tokens the model is deciding between for each token. In this case, baseline perplexity is about 45, and conflicting is around 60. The results in Table 4 are for all tokens in the dev set, whereas the QA results from each epoch in Table 5 show accuracy and F1-macro score on only the first 500 examples in the dataset to avoid overhead in computation.

After completion of training, we load the trained models from their last checkpoints. Then, we perform full inference on the curated QA dataset. For the final stage, we also include the **off-the-shelf** experiment. In total, we have 14,540 questions, distributed across topics. The 5 most common topics and their accuracy across all systems are shown in Figure 1.

Epoch	Eval Loss ↓		Masked Acc. ↑	
	Baseline	Conflicting	Baseline	Conflicting
1	1.7133	2.1003	0.6749	0.6323
3	1.4494	1.8316	0.7172	0.6669
5	1.4014	1.7278	0.7238	0.6778
8	1.3567	1.6925	0.7301	0.6824

Table 4: Dev loss and masked token accuracy across epochs. Lower loss and higher masked accuracy indicate better language modeling performance.

Epoch	Accuracy ↑		F1-macro ↑	
	Baseline	Conflicting	Baseline	Conflicting
1	0.2860	0.2940	0.2851	0.2922
3	0.2980	0.2920	0.2964	0.2894
5	0.3040	0.2980	0.3032	0.2965
8	0.3080	0.2920	0.3079	0.2903

Table 5: Train QA performance across epochs. Higher accuracy and macro-F1 indicate better downstream task performance. Sample of **500** examples.

Table 6 shows the F1-macro score and the overall accuracy on the data. Table 7 demonstrates the support for each answer option (we always have four options) and the F1-scores of each system for each option. Bolded are the best scores.

6 Discussion

Table 4 shows the evaluation cross-entropy loss (i.e. model surprisal) and the masked token accuracy on the dev dataset. We do not include all 8 epochs for readability, but the trend remains similar. The conflicting model tends to be more surprised, on average, on the dev set than the baseline. This might hint that the conflicting information, although not much, has some effect. Both losses decrease gradually throughout the epochs, suggesting the model benefits from more iterations. Looking at the masked accuracy, these statements are confirmed once again. This shows that after the first epoch, the baseline model achieves 67.49% accuracy, compared to only 63.23% for the one augmented with conflicts.

On the other hand, such results are not so clearly visible in the QA task shown in Table 5. This suggests the QA task signal might be too low due to the MLM head and the evaluation pipeline in general. Comparing both accuracy and F1-macro scores, they are initially almost equal across both experiments, but the baseline model is always slightly better. It is important to keep in mind that only 500 examples are evaluated per epoch, as this process

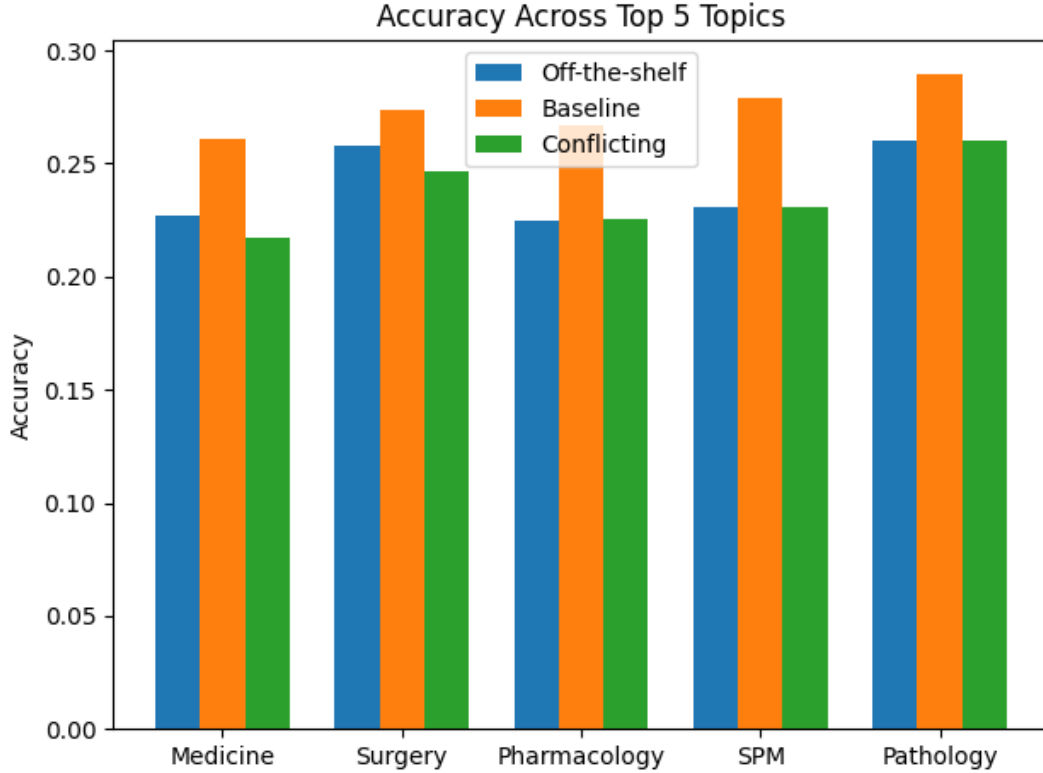


Figure 1: Accuracy across the top 5 most frequent topics for Off-the-shelf, Baseline, and Conflicting models.

Model	F1-macro $\uparrow$	Accuracy $\uparrow$
Baseline	0.2776	0.2790
Off-the-shelf	0.2409	0.2387
Conflicting	0.2358	0.2341

Table 6: QA performance comparison.

Model	opa (4400)	opb (3695)	opc (3310)	opd (3135)
Baseline	0.3062	0.2713	0.2609	0.2719
Off-the-shelf	0.2243	0.2274	0.2456	0.2665
Conflicting	0.2255	0.2206	0.2394	0.2576

Table 7: Per-class F1-scores. Additionally, with support indicated in parentheses.

repeats.

However, the differences become more pronounced at full inference time. Figure 1 clearly shows that across the five most common topics in the data, the baseline model is superior to its counterparts. This trend holds for almost all other topics as well. Another interesting finding is that the off-the-shelf model is generally equal to or slightly better than the conflicting one. If anything, this is at least a good indicator that our domain adaptation approach does not significantly harm the

model’s general knowledge. Looking at Table 6, the baseline model shows more visible differences than the off-the-shelf and conflicting ones. Its final F1-macro score is 27.76%, which is more than 3 points than the off-the-shelf score. The scores of off-the-shelf and conflicting models remain rather similar. For accuracy, the observed behaviour repeats. Finally, Table 7 shows the distribution of answer options and the model’s F1-scores. The most common option choice is (a), for which our baseline model has the highest score. This time, the difference between the baseline and the conflicting model is about 8%. Overall, the baseline model performs best across all options, suggesting it is robust even when the original questions are paraphrased.

7 Conclusion

We investigated the effects of adapting the efficient Transformer-based BigBird to a target domain, which was sometimes purposefully augmented with conflicting contexts. More specifically, we developed a conflict planting framework, a paraphrasing strategy, and an evaluation pipeline alongside our training methods to tailor BigBird’s behaviour for our use case.

From our experimental setup, we find that the baseline model (i.e. the one trained on the human-provided contexts) performs notably better than the model with augmented conflicts. This is confirmed both during training, through measuring cross-entropy loss and masked token predictions, and during inference with our paraphrase version QA dataset.

Finally, we show that even intra-memory conflicts introduced during training lead to increased uncertainty and consistent performance degradation during inference time. Our approach highlights that targeted data perturbations provide principled way to probe how conflicting knowledge is encoded, which is essential for improving the reliability of LLMs in high-stakes medical domain.

8 Future Work

We see potential in extending this work in the future. We believe more interesting findings regarding the rise of conflicts during pretraining and their effects on downstream NLP tasks are to be explored. Moreover, work on expanding relevant corpus and adopting more fine-grained evaluation approaches pose promising future directions.

Extensive and more fine-grained analysis could be achieved by planting the conflict only for specific topics. This would allow us to see whether conflicting subjects influence others.

Additionally, we could explore the influence of fine-tuning on QA by training a classification head to evaluate the model’s conflict handling capabilities in a down-stream task.

One further aim may be to assess the model’s ability to distinguish between factually correct and incorrect conflicting cases. This could be achieved by experimenting with training objectives and decoding strategies, e.g. by using contrastive loss in addition. We perceive our work as valuable contribution to relevant advancements in the study of intra-memory knowledge conflicts.

9 Limitations

It is important to outline the limitations of this research project. Starting off with the size of the available corpora. Due to time constraints, filtering of faulty cases, and balancing the topic distributions, our final training data size is 3M tokens. Some domain-specific terminology may be excluded to achieve good performance on prominent benchmarks, such as MedMCQA. Moreover, multiple

experimental runs would ensure reproducibility of our results.

Furthermore, it is important to note the nature of the domain adaptation dataset. Comprising mainly concatenated short contexts discussing a wide variety of medical topics, such data may easily “confuse” the model, leading to significant performance degradations if DA is performed aggressively. Usually, datasets for such tasks are scraped from longer, more coherent sources.

Another possible limitation is the way our evaluation is performed. A potential issue is that the model might favour options with common word order patterns more, even if they are entirely unrelated to the question. We partially mitigate the bias long sequences introduce in the overall likelihood of the sequence by normalising the scores by length, but we keep in mind that this might not be enough. To confirm this, however, a deeper investigation is needed.

10 Ethical Considerations

The MedMCQA dataset used for augmentation and conflict-planting is human-generated but contains a series of entries the model considers sensitive or violent. Such cases were omitted.

Generation of CLuster, the server belonging to the Institute of Computational Linguistics at the University of Heidelberg.

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We use ChatGPT and Perplexity partially for research, formatting, as well as LaTeX expressions and citations. Additionally, we use Grammarly for help with spelling and phrasing.

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A Appendix A

A.1 MedMCQA

There are two types of entries: single- and multiple-choice. Technically, the latter are still single-choice, since there’s always only one correct option. The difference lies in the wording of the question. Standard questions allow for four options. For cases requiring a longer list or asking about the correct order, another level of abstraction is added: options are listed directly after the question, becoming part of it, while the answer options feature different combinations of letters referencing them. Examples can be seen in 1 and 8.

Question: Characteristics of Remifentanyl – a) Metabolised by plasma **esterase**) Short **half life**) More potent than **Alfentanyl**) Dose reduced in hepatic and renal **disease**) Duration of action more than Alfentanyl

Cleaned Question: Which of the following are characteristics of remifentanyl? a) Metabolised by plasma esterase b) Short **half life** c) More potent than Alfentanyl d) Dose reduced in hepatic and renal disease e) Duration of action more than Alfentanyl

Options: A. ab B. bc C. abc D. bcd

Explanation: Remifentanyl is the shortest acting opioid due to its metabolism by plasma esterase → dose adjustment is not needed in liver or kidney disease. It is more potent than **alfentanyl** : Order of potency is Sufentanyl > Fentanyl = Remifentanyl > Alfentanyl.

Cleaned Explanation: Remifentanyl is the shortest-acting opioid due to its metabolism by plasma esterase. This characteristic eliminates the need for dose adjustment in patients with liver or kidney disease. In terms of potency, remifentanyl is more potent than alfentanyl, and its order of potency is as follows: Sufentanyl > Fentanyl = Remifentanyl > Alfentanyl.

Correct Option: C

Subject: Physiology

ID: 73515f05-e947-4801-8077-3abdeca95c84

Table 8: MedMCQA multiple-choice question example. The issues are coloured in orange.

B Appendix B

B.1 Domain Adaptation

Table 10 shows the exact hyperparameters used for the DA of BigBird. We adopt a more subtle approach by linearly scheduling the learning rate to prevent overly negative impacts on general knowledge. This setting is used across the **baseline** and **conflicting** model setups.

C Appendix C

C.1 Paraphrasing Prompt Engineering

The prompt that yielded the highest-performing paraphrases is shown in Figure 2.

With these instructions, the model performed its task fairly faithfully, requiring minimal, if any, supervision compared to options which used less authoritative or absolute language. Even though the 3B checkpoint was smaller than others we tried, the instructions above resulted in fewer hallucinations than larger models with less restrictive language. Ultimately, the model parameters remained unchanged to achieve good performance in the default setup.

C.2 Paraphrasing Quality Assessment

To ensure the quality of our generated data, we deployed several automated metrics during and after generation. Besides improving overall quality, this would also prevent the use of malformed paraphrases later on. Our automated metrics were BERTScore (Zhang et al., 2020), METEOR (Banerjee and Lavie, 2005), and Rouge (Lin, 2004). BERTScore is an embedding-focused metric that scores alignment at the sense and embedding layers between the original question and the paraphrased sentence. It is known to overestimate sentence closeness; thus, we opt for METEOR as an additional metric. METEOR, which is based on the BLEU (Papineni et al., 2002) score, measuring only exact word overlap, also accounts for synonyms, stem matches and word order differences. The third and final of our metrics is Rouge. Rouge focuses on n-gram overlap, which is interesting for comparing the counts of individual n-grams in the originals that are also found in the paraphrases. Results in the Table 9 show that the paraphrasing remained faithful while generally not reusing a lot of the original sentences’ words. A Rouge-1 score of 0.3 points at only a third of the original questions’ words found in the paraphrase, whereas Rouge-2 denotes that only about 10% of the original words made it into the paraphrase as a pair. The higher Rouge-L score hints at **something**. METEOR stays consistent at the 0.33-0.36 range. This shows that even though the individual-word overlap might not be high, the paraphrases do not have an overly large number of synonyms. BERTScore remaining around 0.9 shows that, even though the questions do not have much overlap in reused vocabulary, they remain topically and sensibly con-

sistent, allowing the paraphrases to remain faithful to the original while adopting different structures and words.

Split	R-1	R-2	R-L	M	B F1
Train	0.357	0.138	0.291	0.338	0.897
Valid	0.369	0.148	0.301	0.349	0.901
Test	0.357	0.133	0.288	0.361	0.897

Table 9: Average evaluation metrics across dataset splits. R-1: ROUGE-1, R-2: ROUGE-2, R-L: ROUGE-L, M: METEOR, B F1: BERTScore F1

C.3 Paraphrasing Error Analysis

A lot of the errors which had surfaced during the generation were of two natures: (1) shift in meaning and (2) grammatical issues.

While the metrics support the claim that the model has used a wide range of differing semantic structures, more often than not, faulty generations were spotted due to wrongly structured sentences. These sentences were often still grammatically correct and well phrased, but shifted the topic or focus of the original question, altering its meaning. In such cases, the model would be reprompted until it produced a topically correct output. While grammatical errors were less common, if the sentence was indeed malformed, the model would be reprompted until it was grammatically and topically correct. To measure these, the three aforementioned metrics were again used. This automated process would be guided by the following rules: 1. If the output was below 0.2 in either METEOR or Rouge score, the original question as well as the generated output would be automatically reprompted. This threshold was chosen due to the fact that most generations going below 0.2 METEOR or Rouge-1 were either malformed or hallucinated. If the BERTScore, due to its tendency to overestimate topic similarity, fell below 0.6, the offending question and generation would be written to a separate file and manually evaluated. This evaluation would be grounded in the inspecting human’s personal judgment of whether this generation is still valid or not. The process was possible because the BERTScore was rare to fall below 0.6. This manual inspection allowed for a couple of generations to be salvaged and many more to be redone.

You are a paraphrasing expert who does NOT under any circumstances answer questions but paraphrases them instead. Do not change the meaning. You may use synonyms or change the sentence structure. You may shorten or lengthen the question, but you are incapable of answering it. You are a paraphrasing expert. NEVER answer questions. ONLY output 3-5 rephrased versions of the input question. Number them 1., 2., etc. Keep meaning identical.

Figure 2: Question paraphrasing system prompt

Parameter	Value
Model	google/bigbird-roberta-base
MLM Probability	0.15
Learning Rate	3e-5
Warmup Ratio	0.05
Weight Decay	0.01
Max Gradient Norm	1.0
Max Sequence Length	2048
Number of Epochs	8
Dataloader Workers	4
Train Batch Size (per device)	4
Eval Batch Size (per device)	2
Gradient Accumulation Steps	2

Table 10: Hyperparameters for MLM Domain Adaptation

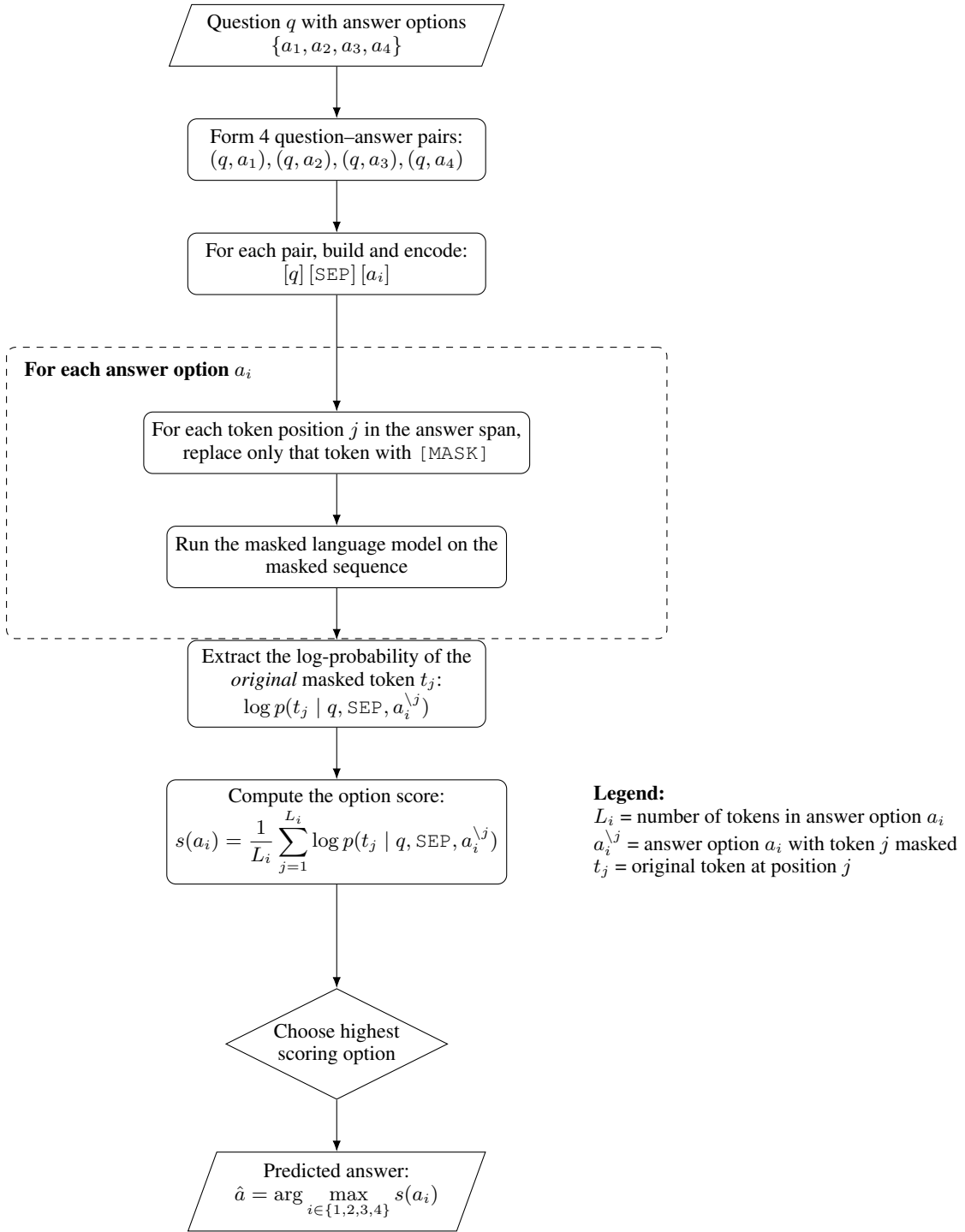


Figure 3: Normalised-likelihood scoring of answer options with our domain-adapted models. For each question, each answer option is appended to the question using a [SEP] token. Each token in the answer option is masked one at a time, and the model assigns a log-probability to the original masked token. The normalised sum of these log-probabilities is used as the option score, and the highest-scoring option is selected.